

Guideline Title: Inhalation Therapy

Guideline Number: GUID-3203-PR012

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To optimize oxygenation, prevent or reverse hypoxia in acute bronchospasm, and decrease airway swelling and or stridor in cases of reversible non-foreign body upper airway obstruction.
- II. **DEFINITIONS:** None
- III. **DEPARTMENT GUIDELINE:**
 - A. **EQUIPMENT:**
 - i. Small volume nebulizer
 - ii. Oxygen source
 - iii. Albuterol in 0.083% pre-mixed solution
 - iv. Ipratropium (Atrovent) 0.5mg solution
 - v. Vaponephrine (racemic epinephrine) in 2.25% solution
 - B. **INDICATIONS:**
 - i. Bronchospasm
 - ii. Partial airway obstruction caused by croup, epiglottitis or laryngospasm
 - iii. Hyperkalemia
 - C. **CONTRAINDICATIONS:**
 - i. Use caution in patients with tachycardia-related cardiac dysrhythmias
 - ii. Use with caution in patients with known coronary disease
 - iii. Consider termination of treatment if heart rate increases more than 20%
 - iv. Do not use Atrovent with patients who have a known history of a Peanut allergy
 - D. **General Assessment:**
 - i. Use appropriate PPE for the patient's condition
 - ii. Assess the adequacy of the airway, breathing and circulation, intervene as appropriate
 - iii. Provide ventilatory support as needed
 - iv. Monitor and record vital signs including RR, SPO2, HR, ECG and BP
 - v. Administer high flow, high concentration oxygen
 - E. **PROCEDURE: Adult/pediatric General - nebulizer**
 - i. Prepare the nebulizer with medication per condition below
 - ii. Connect the nebulizer to an oxygen source at 8 lpm

Guideline Title: Inhalation Therapy

Guideline Number: GUID-3203-PR012

- iii. Instruct the patient to seal their lips around the mouthpiece of the nebulizer and inhale through their mouth deeply
 - iv. If the patient is unable to seal their lips around the mouthpiece the treatment may be given as blow-by or connected to a mask
 - F. **Condition:** Adult/pediatric bronchospasm
 - i. Prepare the nebulizer with
 - 1. Adult: **Albuterol 2.5 mg/3 mL**
 - 2. Pediatric: **> 12 years old: Albuterol 2.5 mg/3 mL q 20 min** may repeat x3. If needed, continuous nebulizer dose: 10-15 mg/h
 - 3. Pediatric: **< 12 years old: Albuterol 0.15-0.3 mg/kg q 20 min** may repeat x3. If needed, continuous nebulizer dose: 0.5 mg/kg/h
 - 4. **Ipratropium bromide 0.02% 0.5 mg (2.5 mL)** to the initial albuterol nebulized x 1 dose.
 - G. **Condition:** Partial airway obstruction due to croup, epiglottitis, or laryngospasm.
 - i. After obtaining an order medication from the sending physician, prepare the nebulizer with **Racemic epinephrine 0.05 mL/kg** of a 2.25% solution to a max single dose of 0.5 mL and may repeat every 20 minutes prn.
 - a. **Condition:** Severe ARDS
 - i. If patient is being administered aerosolized **Epoprosterol (Flolan)** and the appropriate equipment is available, continue at current dosing in consultation with sending/admitting physician or Medical Control
 - b. **Condition:** Hyperkalemia
 - i. Prepare the nebulizer with
 - 1. Consider **Albuterol 2.5 mg/3 mL** of NS via nebulizer, may repeat.
- IV. **DOCUMENTATION:**
- A. EMS Charts
- V. **REFERENCES:**
- A. None